Native Binary-Weight Training Mitigates Post-Training Quantization Collapse in MNIST Diffusion

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***Abstract*—Does failure after projecting a diffusion model to one-bit weights show that its binary representation is unusable? We test that implication, rather than propose a new quantizer, in a parameter-compatible MNIST residual U-Net. We compare floating-point training, zero-shot one-bit-weight post-training quantization (PTQ), binary-weight training from random initialization, floating-point warm-started quantization-aware training (QAT), and a binary-weight/ binary-activation core. Every condition uses the same topology, data, optimizer, per-stage budget, and 1,000-step ancestral sampler, with three independent training seeds. Native W1A32 training obtains domain-feature Fréchet distance 46.33 ± 17.24, 28.4-fold lower than matched zero-shot PTQ (1314.20 ± 314.96), but remains substantially worse than FP32 (3.15 ± 0.38). Warm QAT also recovers much of the PTQ loss, while the W1A1 core retains a large distributional gap. A simple constrained- optimization proposition formalizes why one projected floating-point optimum is not an impossibility certificate for the binary hypothesis class. The supported claim is deliberately local: on MNIST, training path explains a substantial part of one-bit-weight PTQ collapse; this does not establish full W1A1 success, floating-point parity, or transfer to natural images.**

***Index Terms*—binary neural networks, diffusion models, post-training quantization, quantization-aware training, MNIST**

I. Introduction

Diffusion models repeatedly evaluate a time-conditioned denoiser, making model memory and arithmetic attractive compression targets. Binary neural networks replace many real-valued weights by signs and optional scales [1], [2]. At the extreme, BiDM studies binary weights and activations (W1A1) and motivates specialized architecture and distillation by the severe degradation of naive full binarization [3]. BinaryDM studies accurate weight binarization at W1A4 [4].

A narrower logical question precedes those system designs: if a trained FP32 denoiser collapses after sign projection, has the binary-weight representation failed, or has a point optimized for a different forward model simply been projected poorly? Zero-shot PTQ evaluates only the projected FP32 solution. BinaryConnect-style training instead maintains latent real parameters while using binary weights in every forward pass [5]. The two procedures need not reach the same binary point.

We isolate this distinction on MNIST. All precision conditions share parameter names, shapes, topology, optimizer, sampling procedure, and evaluation. The experiment supports three findings:

- native W1A32 training reduces dFID by 28.4-fold relative to matched zero-shot PTQ, so PTQ collapse is not a certificate of binary-weight representational failure in this setting;
- warm-started QAT also recovers substantially, showing that random initialization is not uniquely necessary once binary forwards are optimized; and
- FP32 remains much better, while a W1A1 core remains weak; the result therefore neither contradicts the difficulty identified by BiDM nor establishes a general binary-diffusion solution.

The contribution is a controlled empirical distinction between projection and optimization. The sign/scale quantizer, straight-through estimator (STE), and residual components are standard and are not claimed as algorithmic novelty.

II. Why PTQ Failure Is Not an Impossibility Result

Let Θ_F be a floating-point parameter space, Θ_B the feasible binary-weight space, $\mathcal{L}$ the diffusion training objective, and $P : \Theta_F \to \Theta_B$ the sign/scale projection used by PTQ. Let $\theta_F^\star$ minimize $\mathcal{L}$ over Θ_F and $\theta_B^\star$ minimize it over Θ_B.

Proposition 1 (Projection non-implication)**.** *The loss of a projected floating-point optimum is an upper bound on the optimal binary-constrained loss:*

$$\mathcal{L}(\theta_B^\star) \le \mathcal{L}(P(\theta_F^\star)). \tag{1}$$

Consequently, poor performance of $P(\theta_F^\star)$ alone cannot establish that every member of Θ_B performs poorly.

Proof. Because $P(\theta_F^\star) \in \Theta_B$, it is one feasible point in the minimization defining $\theta_B^\star$. The minimum over Θ_B cannot have higher loss than that particular feasible point. □

This elementary statement is not a convergence result. Native STE training is not guaranteed to find $\theta_B^\star$, and diffusion quality is not determined by training loss alone. It identifies exactly what the experiment must test: whether optimization under binary forwards reaches a materially better feasible point than zero-shot projection.

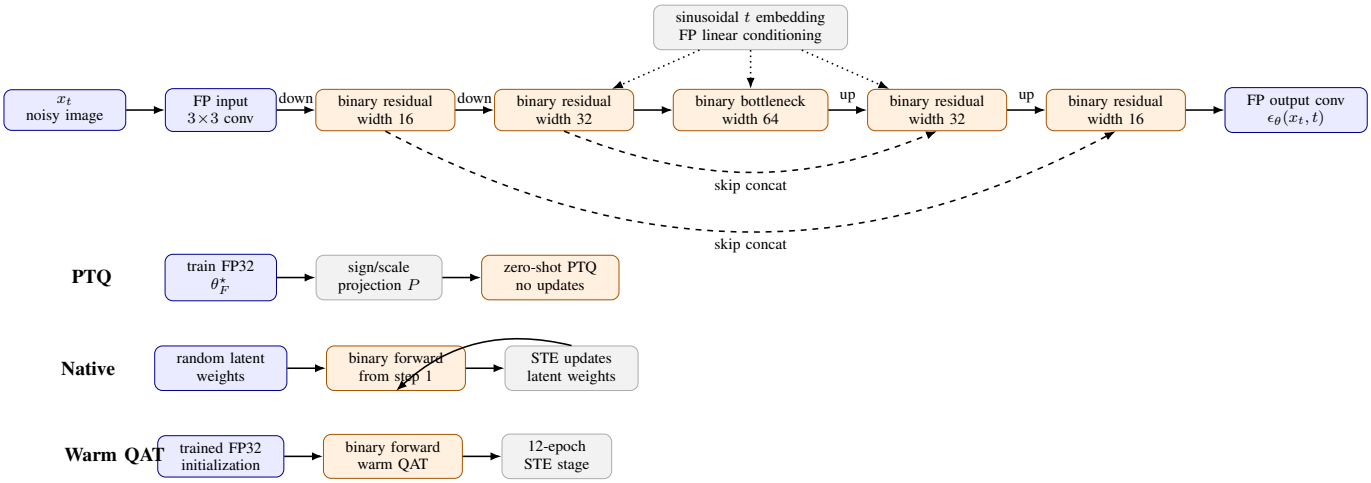


Fig. 1: Top: compact residual U-Net used in every condition. Binary-marked convolutions use W1A32 unless the W1A1 core is selected; normalization, conditioning, scales, residual additions, and boundaries remain FP32. Bottom: the controlled intervention changes the optimization path, not the topology.

III. Controlled MNIST Experiment

A. Denoiser and binary forward

For clean image x_0, noise ϵ, and timestep t, every condition minimizes the DDPM noise-prediction objective [6]:

$$\mathbb{E}_{x_0,t,\epsilon} \left\| \epsilon - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t \right) \right\|_2^2 . \tag{2}$$

The denoiser in Fig. 1 is a residual U-Net [7] with base width 16, widths $[16, 32, 64]$, two downsampling stages, two skip concatenations, GroupNorm, and sinusoidal time embeddings [8]. It has 218,993 trainable parameters.

For output filter W_o, the primary uncentered binary projection is

$$\widehat{W}_o = \alpha_o \operatorname{sign}(W_o), \qquad \alpha_o = \operatorname{mean}(|W_o|). \tag{3}$$

Latent FP32 parameters receive identity-STE gradients [9]. The centered ablation subtracts the filter mean before sign and scale. The W1A1 core additionally signs block activations with a clipped-identity backward pass. Input/output convolutions, normalization, time conditioning, biases, scales, accumulation, and residual additions remain FP32; "W1A1 core" therefore does not mean an end-to-end binary appliance.

B. Conditions and optimization

FP32 and native conditions train from random initialization for 12 epochs. PTQ loads the corresponding completed FP32 checkpoint, changes the forward projection, and performs no optimization. Warm QAT loads that checkpoint and adds a full 12-epoch binary-forward stage, for 24 total epochs. Native W1A32 crosses centered/uncentered projection with pre-/post-activation residual ordering [10]; the headline matched comparison is uncentered and pre-activation.

We use the canonical 60,000/10,000 MNIST split [11], scale images to $[-1, 1]$, and train with AdamW [12], learning rate 2×10^{-4}, weight decay 10^{-4}, cosine decay, gradient clipping at 1, batch size 256, and training seeds $\{0, 1, 2\}$. Sampling uses all 1,000 ancestral DDPM steps. Each reported distribution contains 2,000 samples from a fixed sampling seed independent of the training seed.

C. Evaluation and controls

An independently initialized MNIST classifier, trained only on the real training split, reaches 98.03% test accuracy and 1.44% ten-bin expected calibration error. Its penultimate features define domain-feature Fréchet distance (dFID) [13], unbiased polynomial-kernel distance (dKID) [14], and $k = 5$ density and coverage [15]. We also report Jensen–Shannon divergence of predicted class priors. These domain-specific values are not comparable to ImageNet FID.

A disjoint real–real split gives dFID 0.076, density 0.969, and coverage 0.971. Clamped Gaussian noise gives dFID 2278.24 and zero density/coverage. Although that noise receives mean peak classifier confidence 1.000, confidence is not used as a quality metric.

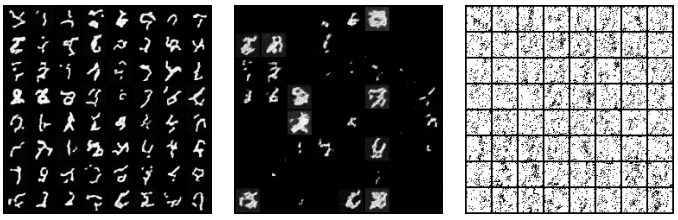
IV. Results

a) Native optimization mitigates projection collapse.: Table I gives the main comparison. Native W1A32 has 28.4-fold lower mean dFID than its matched PTQ configuration and nonzero density and coverage. This is an empirical witness that a binary-forward optimization path can reach a substantially better model than sign-projecting the FP32 endpoint. It does not show that all binary training succeeds: the standard deviation is large and every reported distribution metric remains worse than FP32.

b) Warm QAT also recovers.: Warm QAT obtains dFID 66.45 ± 28.49. Its centered configuration is not a strictly matched comparison to the best uncentered native row, but its recovery from PTQ demonstrates that FP initialization is not intrinsically harmful once the projected model is optimized. Native training is superior to zero-shot PTQ, not necessarily to every warm-start or teacher-dependent procedure.

TABLE I: MNIST metrics, mean ± sample standard deviation over three training seeds. Each seed contributes 2,000 samples from the same ancestral sampler. Bold marks the best non-FP condition, not parity with FP32.

Condition	dFID ↓	dKID ↓	Density ↑	Coverage ↑	Class JS ↓
FP32	3.15 ± 0.38	0.056 ± 0.013	0.284 ± 0.014	0.195 ± 0.017	0.033 ± 0.009
Native W1A32, uncentered/pre	$\mathbf{46.33 \pm 17.24}$	$\mathbf{1.286 \pm 0.816}$	$\mathbf{0.022 \pm 0.005}$	$\mathbf{0.033 \pm 0.010}$	$\mathbf{0.295 \pm 0.020}$
Warm QAT W1A32, centered/pre	66.45 ± 28.49	4.945 ± 3.686	0.019 ± 0.004	0.032 ± 0.004	0.320 ± 0.059
Native W1A1 core	204.55 ± 69.77	41.03 ± 25.01	0.005 ± 0.002	0.009 ± 0.004	0.339 ± 0.027
PTQ W1A32, uncentered/pre	1314.20 ± 314.96	2467.50 ± 1549.51	0	0	0.531 ± 0.002



(a) FP32 (b) Native W1A32 (c) PTQ W1A32

Fig. 2: Fixed-seed samples. Native binary-weight training retains digit-like structure but remains visibly below FP32; zero-shot PTQ collapses.

TABLE II: Native W1A32 architectural ablation (mean dFID; lower is better).

Residual order	Uncentered	Centered
Pre-activation	46.33	99.50
Post-activation	84.26	82.46

c) W1A1 remains difficult.: The W1A1 core improves over zero-shot W1A32 PTQ in dFID but has density 0.005 and coverage 0.009. Our experiment therefore does not refute BiDM's observation that simultaneous weight and activation binarization is difficult. It instead isolates a weaker but useful statement about one-bit weights with FP32 activations.

d) Architecture does not yield a universal rule.: Table II shows an interaction: pre-activation helps only without centering. Centering also raises measured latent-to-projected angular error from 33.29° to 34.05° in the pre-activation pair. These results reject our preliminary centering explanation and do not support a universal claim that either centering or residual ordering prevents collapse.

V. Scope, Limitations, and Reproducibility

This is an MNIST existence result for a small unconditional model. Three seeds expose high variance but are insufficient for fine-grained significance claims; 2,000 samples per seed make absolute metric estimates noisier than common 50,000-sample evaluations. A separate compact CIFAR-10 attempt is intentionally excluded from the evidence because its FP32 control also failed, so it could not adjudicate binary versus floating-point quality on natural images. No theorem here establishes STE convergence, score approximation, or scaling behavior.

Binary convolutions contain 195,328 of 218,993 parameters. Physical sign packing plus FP32 scales and remaining tensors gives a 121,316-byte tensor payload versus 875,972 bytes for FP32, a $7.22\times$ whole-model reduction. This is a storage measurement: ordinary framework convolutions still execute floating-point operations, and we claim no latency or energy improvement.

The released pipeline [16] records configurations, seeds, environment versions, source commit, checkpoint SHA-256, histories, metrics, controls, and packed exports. Exact entry points are under `experiments/v1_binary_study`; the reported artifacts are under `artifacts/v1_binary_study/mnist`. Tests cover parameter compatibility, forward shapes, diffusion schedules, metric controls, and physical packing.

VI. Conclusion

Zero-shot projection answers whether one particular FP32 endpoint survives binarization; it does not answer whether binary-forward optimization can find a better binary-weight solution. In a controlled MNIST diffusion model, native W1A32 training reduces domain-feature distance by 28.4-fold relative to matched PTQ, while warm QAT also recovers substantially. The result is meaningful but bounded: FP32 remains much better, W1A1 remains weak, and natural-image transfer is unresolved. Future binary-diffusion comparisons should distinguish native, warm-started, distilled, and zero-shot training paths rather than treating PTQ collapse as the limit of the representation.

Code availability. Source, checkpoints, packed weights, and evaluation artifacts are available at https://github.com/elixpo/monodiffuse.

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